



Feasibility Report

Version-1.0

System Analysis, Design & Development Sessional [CSE 402]

Project: Medica

Group B2

Date: April 30, 2021

SL	ID	Name	Email
1	201814015	Dhrubo Barua	iamdhrubobarua@gmail.com
2	201814021	Md. Rokonuzzaman Reza	rokonuzzamanreza@gmail.com
3	201814023	Ishraq Hasan	ishraq10199@gmail.com
4	201814044	Kazi Tasnim Rahman	tasnim.naomi@gmail.com
5	201814052	Muhammad Munswarim Khan	munswarim@gmail.com
6	201814059	Shaqran Bin Saleh	shaqran39@gmail.com

Table of Content

Abstract/Executive Summary	3
Revisions	4
1. Introduction	5
2. Existing System	6
2.1. Information Gathering and Analysis	6
2.1.1. Overview and Objective	6
2.1.2. Information required for system analysis	6
2.1.3. The sources of information	7
2.1.4. Information gathering methodology	7
2.2. Presenting the Existing System	7
3. Proposed Candidate Systems	10
3.1. Potential Candidate Systems	10
3.2. Characteristics of the Candidate Systems	11
3.3. Performance and Cost-effectiveness of the Candidate Systems	12
3.4. Selection of the Best Candidate Systems	13
4. Specifications of the Selected Candidate Systems	14
5. Conclusions	18
6. Appendices	18
Appendix-A	18
Appendix-B	18

Abstract/Executive Summary

The lack of immediate access to data regarding medical services of caregivers and organizations such as hospitals, and the absence of performance benchmarking in the field of medical practice leads to poor healthcare services throughout the country. Patients are not equipped with enough information about where to look and are mostly unaware of whether or not a better alternative exists. This project aims to identify the aspects of weak infrastructure in the fragmented medical community and propose a system to solve them using modern database management techniques.

There's lack of immediate access to data regarding medical services leading to poor healthcare services throughout the country. Patients struggle to find proper information regarding medical facilities. This leads to a weak network of information among the general populace. Rumors and referrals from friends and family get trusted, instead of having a solid source of data. This has been a source of many cases of undocumented malpractice in the country. *

***We aim to provide infrastructure to promote a system for the patients, to review doctors and practitioners, to get information about which services are better, and location-specific recommendations for their needs in case of emergency. This will help the general public to avail a proper source of important information and provoke an incentive among the medical practitioners to improve their services. This if scaled globally will benefit every single people seeking medical aid. ***

***Project Medica is an online portal for all people in the related field of healthcare. Through this portal, patients can find doctors, care-givers easily, and can contact them. Along with the patients, the doctors and service givers are benefitted as they can easily keep track of the patients, their appointments, and also the prescriptions and reports of the patients.

Keywords:

Hospital, Healthcare, Appointment, Centralization, Emergency, Electronic health records

Revisions

Ver. no.	Primary Author(s)	Description of Version	Date of Completion
0.0	Ishraq Hasan, Kazi Tasnim Rahman, Muhammad Munswarim khan, Shaqran Bin saleh, naim Ibna Khadem	In this version, we took interviews from the hospitals and designed our database system. It was designed for supporting multiple hospital management systems.	April 27, 2021
0.1	Md.Rokonuzzaman Reza, Ishraq Hasan, Kazi Tansim Rahman, Muhammad Munswarim khan, Shaqran Bin saleh	In this version we proposed different candidate systems, analysed their feasibility and selected the best candidate system for solving the problem	May 19, 2021

1. Introduction

In Bangladesh, the field of medicine and clinical practices are quite developed. But there is yet to be a system to unify many of the facilities available to the public, and index them according to their cost-feasibility and distance. Due to the advent of a very unstable economy, doctors and other medical practitioners have started to prioritize their income, more than the well-being of the patients.

Patients struggle to find proper information regarding medical facilities, and a doctor is very hesitant about referring patients to practitioners due to lack of financial gain thereof. This leads to a very weak network of information among the general populace. Rumors and referrals from friends and family have to be trusted, instead of having a solid source of data. This has been a source of many cases of undocumented malpractice in the country.

Project *Medica* is the attempt at solving issues like this. It aims to provide a proper infrastructure and gateway to promote a system for the patients, to review doctors and medical practitioners, getting information about which services are better, and location-specific recommendations for their needs in case of dire emergency. This will not only help the general public to avail a proper source of important information, but also provoke an incentive among the medical practitioners to improve their services.

2. Existing System

Currently, when a hospital stores health info of a patient in their electronic database, the data is not shared with another hospital and the other hospital has to store the same data for their own. This causes huge redundancy and so it is not a wise approach. Besides, all hospitals do not have their own digital system. Some hospitals do not maintain any website or social media page via which they could share about their offered services. Many hospitals rely on third-party websites to announce their recruitment notices.

2.1. Information Gathering and Analysis

We have gathered some information which will help our project. Some of the niche are given below:

2.1.1. Overview and Objective

We have gathered information about the existing system and also talked and took opinions from doctors, patients and hospitals to know about what different classes of people will be expecting out of our project and to make it even more significant and usable.

2.1.2. Information required for system analysis

- How the hospitals store and manage their system resources and database
- Do hospitals disclose their respective information or patient's personal information to other organizations
- Do hospitals have any recruitment portals?
- Do hospitals have any branding or advertising technique
- Do hospitals follow any systematic way of dispatching ambulances
- Is there any app for doctors to manage their schedules
- Is there any means for hospitals to handle prescription provided by doctors
- How the patients reserve allotments for doctor's consultancy
- Do patients maintain preserve their reports and prescription for referencing in future consultancies
- How does a patient contact with ambulance service providers
- Is it feasible to launch home delivery services of medicines by hospitals and pharmacies
- How do patients look for blood donors and how do blood donors come to know about patients who are in need of blood

- What do patients do when they need nurses' assistance at home
- Is it challenging for nurses or caregivers to attend a patient at home aside their respective job
- How does technicians add data to electronic database management system

2.1.3. The sources of information

The sources of our information gathering can be categorized to the following types:

- Doctors
- Patients/ commoners, with no affiliation to hospitals (other than treatment and diagnosis)
- Hospital's executive member
- Related Online Sources

Information from the first 2 from above was gathered through questionnaires. They were given sets of questions with different options as answers. And the hospital executive members were interviewed at their own office.

2.1.4. Information gathering methodology

The information used two methodologies for information gathering:

- Questionnaires
- Interview

In executing the questionnaires, we opted to the Google forms where people would answer the given questions and a poll gets created to count of the answers we get from the response.

Other than the questionnaires, we did interviews. We had a couple of interview sessions with some executives of different hospitals to get a better understanding about the current status of the hospital systems in our country.

2.2. Presenting the Existing System

At present, the country, even though has been a lot digitized in a lot of fields, has not been digitized in the field of healthcare. The government hospitals along with most private hospitals do not have or maintain their websites and those websites do not provide services. In a country of 3,976 healthcare facilities and hospitals, not more than 3 or 4 provide some services through their website.

But the services they provide are only catered to the patients or people who seek the services of the hospitals. But the doctors and care-giver get no service from this. This is not a case that must be changed. Because the service needs to work both ways.

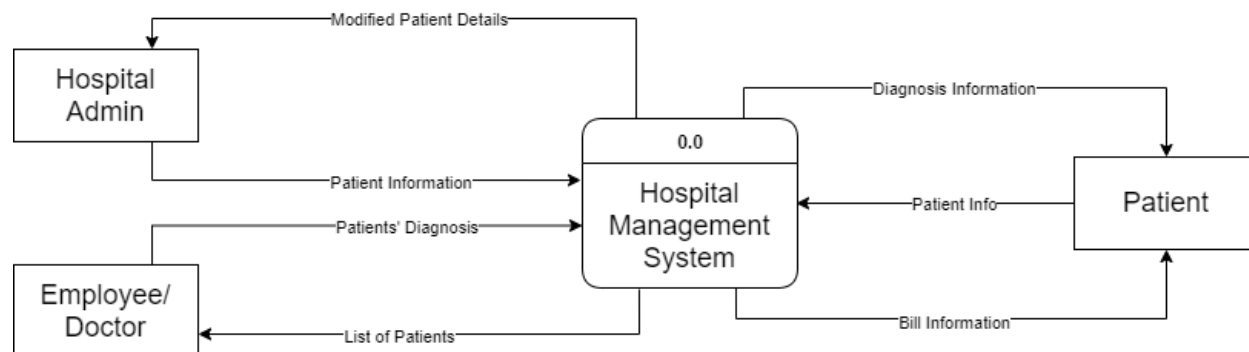


Figure-1: Data flow diagram of a typical hospital management system (level-zero)

And one of the main reasons for the service not being properly introduced is because the hospitals and healthcare-facilities always aim for maximum profit with minimum expenses. And as they do so they find it unnecessary to build up a web-based service provider. Besides because of unskilled personnel, the maintenance is tough and thereby the thought of it is even ignored.

Web-based healthcare service is a very rare phenomenon here in this country and as the country is getting digitized and modernized, the introduction of this service is a must. With this, both doctor and patient along with care-givers will be benefited properly.

There also isn't a systematic approach of finding blood donors during emergency situations. Relatives and affiliates of the patients usually relay the news of urgent need of blood to their familiar ones and they keep looking for donors until the ones willing to donate blood are found.

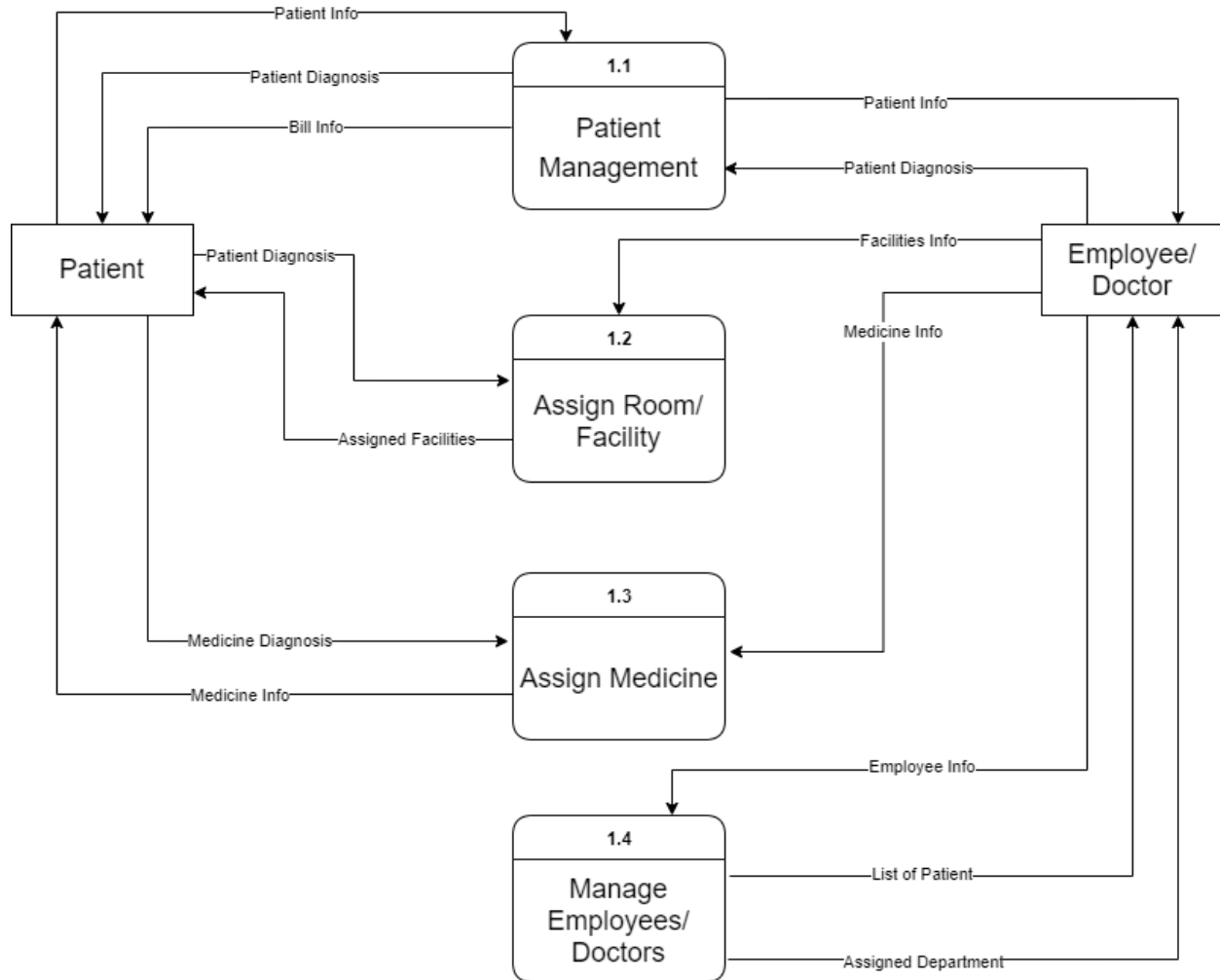


Figure-2: Data flow diagram of a typical hospital management system (level-one)

3. Proposed Candidate Systems

The project aims for a better and easier way of accessing healthcare for the patients along with that it aims for an easier, quicker, and automated version of all the details, reports, prescriptions, and schedules of the patients, and thus, digitalization of the national healthcare system as a whole.

3.1. Potential Candidate Systems

During the corona-pandemic, lots of service-providing organizations, including hospitals, were forced to adopt teleworking and shift towards a digital working environment due to nationwide lockdown. This has given rise to many hospital websites with a motive of fulfilling objectives analogous to that of our project. However, each of these websites are independent of each other, and their database system is not shared or centralized.

Table-1: Features of candidate systems

Candidate Systems	Features
Universal health care system with central administrative system (c1)	• A central administrative system will maintained • Can be used worldwide • All hospitals can share the database • Central ambulance management • Central blood management system
WebRTC based medical consultancy system (c2)	• Doctor and patient can contact in real time • Patients can take healthcare services from their home

Table-2: Hardware and software list for candidate system

Candidates	Hardware	Software
Universal health care system with central administrative system (c1)	Smartphone, Desktop, Laptop	Shared cloud database, Common website, Browser
WebRTC based medical consultancy system (c2)	Smartphone, Desktop, Laptop	Browser, WebRTC based web application

3.2. Characteristics of the Candidate Systems

- Centralization of database system for healthcare management in order to monitor and improve medical services as well as increase its scope
- Modernizing health records to minimize disorientation of data caused by the existing practices of health care management, ultimately, preventing data redundancy
- Navigating/suggesting the quickest and easiest routes/solutions to obtain medical assistance at the time of emergencies
- Digitizing all sorts of formalities required for taking health services
- Providing a user-generated health card which will be linked with respective user's medical profile

Table-3: Comparison between proposed candidate systems

Characteristics	Universal health care system with central administrative system (c1)	sC based medical consultancy system (c2)
Centralized database	A centralized database will be present	No database is present
Security	Doctor's have unique id in the database, no chance of fake doctor	Anyone can pretend to be doctor
Blood donor	A shared system for blood donors, which can notify quickly	No system is present in this
Recruitment portals	There will be a portal for recruitment of doctors, caregivers, technician	There is no portal for recruitment of doctors, caregivers, technicians
Ambulance management	A shared ambulance system will be present	Patients need to search ambulance by own effort
Schedule management app for doctors	An all-purpose applications will be present where doctors and users can make up a profile and manage their essentials	Not required
Digitally prescription management	By scanning patient's personal QR code doctor's can have the access of patients entire previous prescriptions	In this system, a doctor can access patient's medical history without consent

Allotment reservation for consultancy	Required	Required
Home delivery of medicines	Need	Need
Server	Required	High performance, reliable media server required
Internet Connectivity	Only when website is accessed	Uninterrupted connectivity is required for video call
Storage	Cloud	Media server
Website	Required	Required

3.3. Performance and Cost-effectiveness of the Candidate Systems

It takes an annual expense of around **\$10** for owning a server domain name. In addition, a server computer is necessary to host the application and depending on the server's storage, the cost may also vary. In this case, we are suggesting some offers from Amazon. A quad-core server with a RAM of 16GB can be acquired at a monthly expense of **\$61.54** if *Amazon EC2 Instance Savings Plans* instances are picked. Along with this, *Amazon Elastic Block Storage (EBS)*, charging only **\$10** a month, gives a cloud storage of 100GB which can be used to store data. However, the database application needs to be manually installed for this offer. Besides, managing databases of all hospitals centrally will require a large amount of storage and fast transmission techniques. This is where *EBS* may not suffice. If MySQL compatible *Amazon Aurora* is selected, the service will provide 1TB storage along with high bandwidth at a price of **\$160.91** per month. Services may be selected depending on the monetary plan.

Table-4: Candidate Qualitative Evaluation Matrix:

Evaluation Criteria	C1	C2
Performance	Excellent	Good
User acceptability	Excellent	Satisfactory
User responses	Satisfactory	Not ensure
Adaptability	Good	Good
Reliability	Excellent	Poor
Development cost	More expensive	Cheap
Maintenance cost	More expensive	Cheap

Table-5: Weighted Candidate Evaluation Matrix:

Evaluation criteria	Weighing score	C1 rating	C1 score	C2 rating	C2 score
Performance	4	5	20	4	16
User acceptability	5	5	25	3	15
User responses	3	3	9	2	6
Adaptability	3	3	9	3	9
Reliability	4	5	20	1	4
Development cost	3	3	9	4	12
Maintenance cost	4	3	12	4	16
Total			104		78

3.4. Selection of the Best Candidate Systems

From the two candidate systems, we choose the Universal health care system with central administrative system (c1) as the best one. It has better performance, user acceptability and user responses. It is also very user friendly. It also has the highest score in the weighted evaluation matrix which makes it more suitable than the other candidate systems. The reasons of its selection are given below:

- Centralised database will be used
- Portal of recruitment of doctors, caregivers, technician will be present
- A shared ambulance system will be present
- An all-purpose applications will be present where doctors and users can make up a profile and manage their essentials
- Cloud storage will be used

4. Specifications of the Selected Candidate Systems

Functional Requirements:

- Distinguishing UI will be developed for each category of users.
 - Doctors will be provided a UI to maintain their daily schedules, create prescriptions and view patient's medical history.
 - Patients will be provided a UI to view their own medical history, take doctors' appointments, order medicines online, request for ambulances during emergencies, and ask for support in case of blood transfusion.
 - Hospital executives will be provided a UI to maintain their profiles containing a list of services and facilities, as well as use it for administrative management systems and a recruitment portal.
 - Hospital employees like technicians and nurses can access data restricted within their respective hospitals.
 - Caregivers will be provided a UI to maintain their daily commitments and new contracts.
 - Blood donors will be provided notifications when patients of nearby hospitals will require blood transfusion.

Nonfunctional Requirements:

- A secured system will be developed obeying information security policies and protocols
- All the hospitals and clinics will have a common database for patients to reduce data redundancy
- Confidentiality within the certain data within the hospitals will be maintained

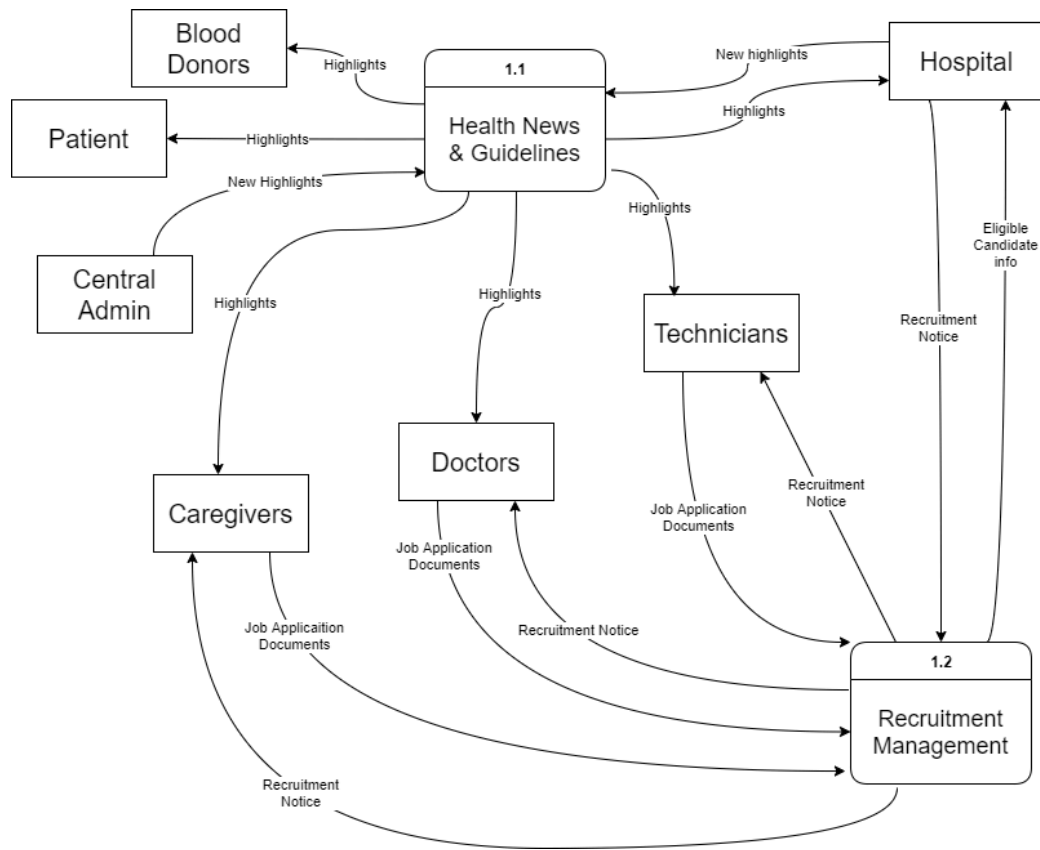


Figure-3: Data Flow diagram for “Notice” component of proposed system

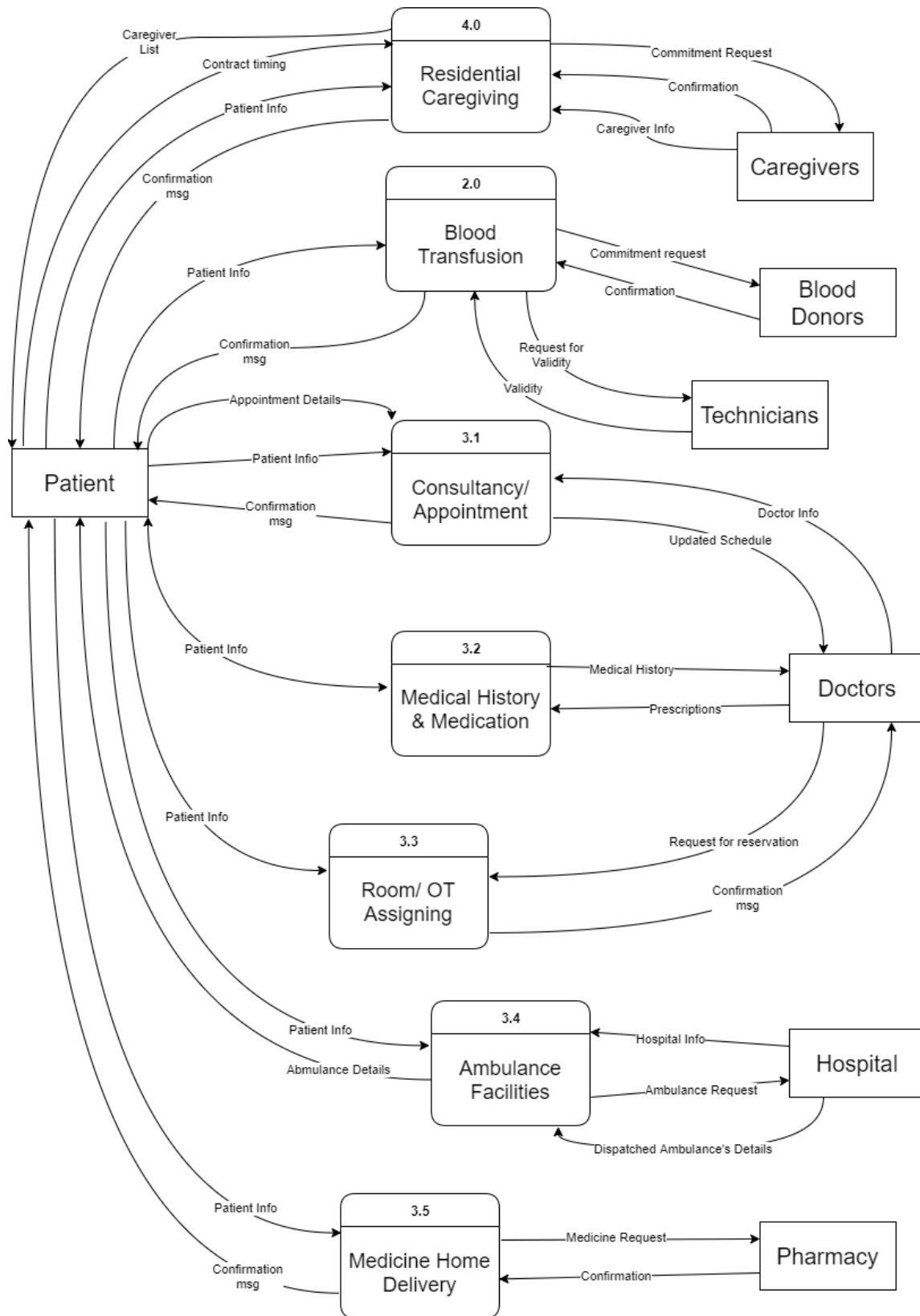


Figure-4: Data Flow Diagram for “Clinical Facilities” component of proposed system

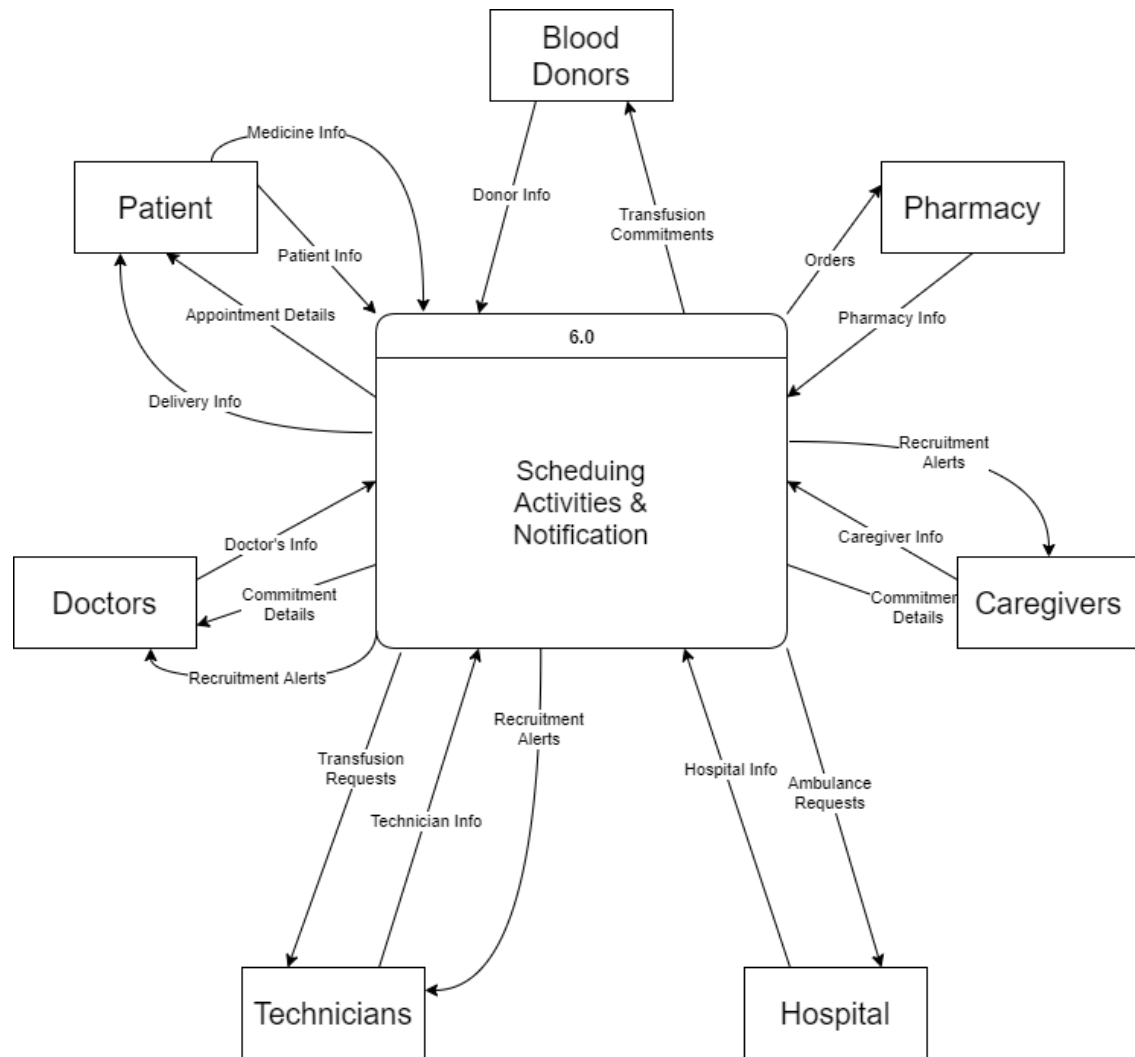


Figure-5: Data Flow Diagram for “Activity Scheduling” component of proposed system

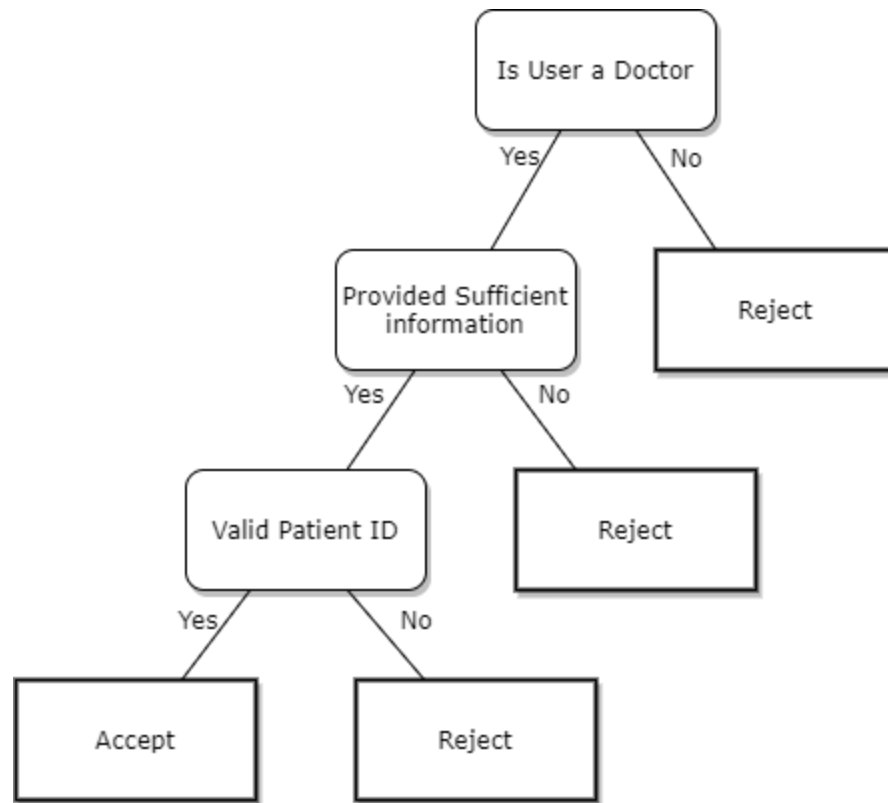


Figure-6: Decision Tree for "Create Prescription" feature

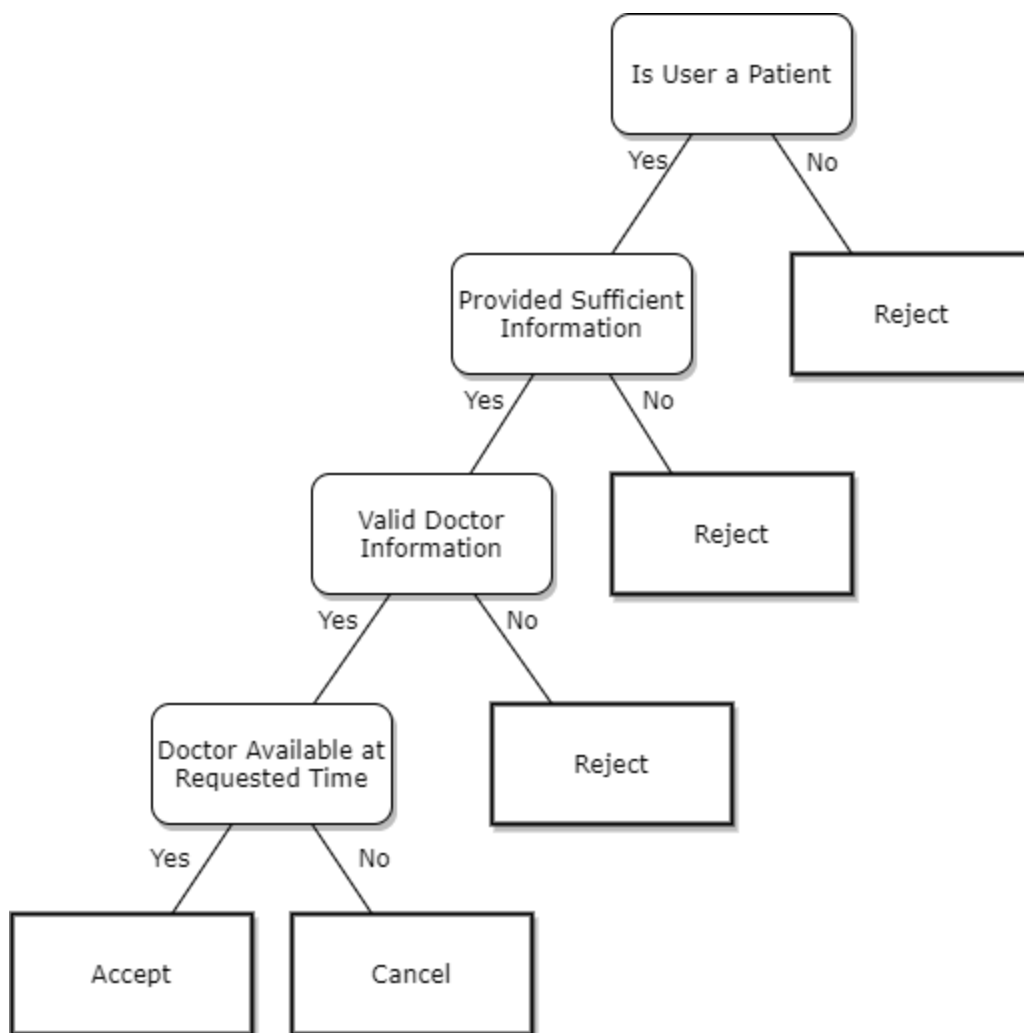


Figure-7: Decision Tree for "Doctor's Appointment" feature

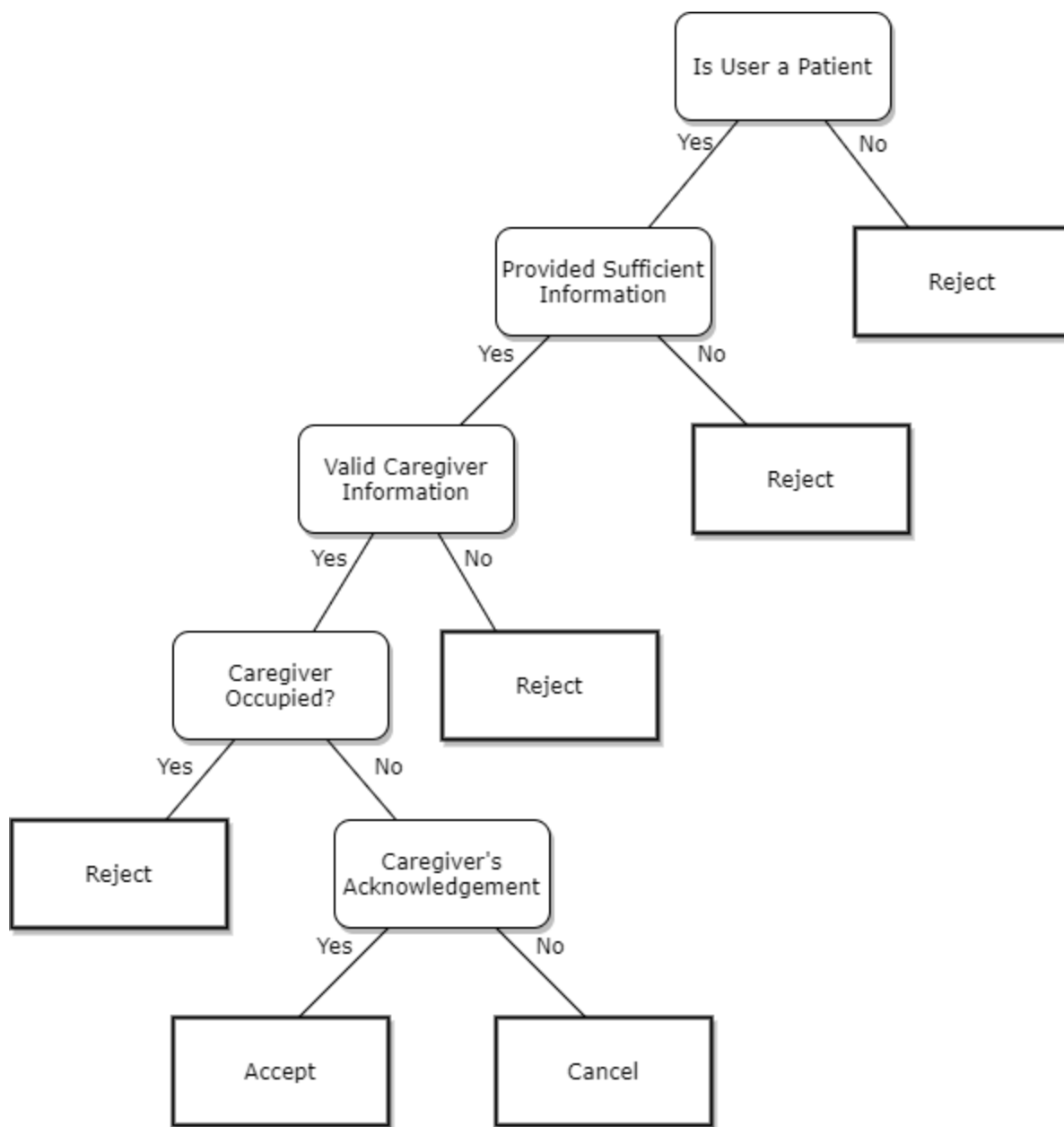


Figure-8: Decision Tree for "Residential Caregiving" feature

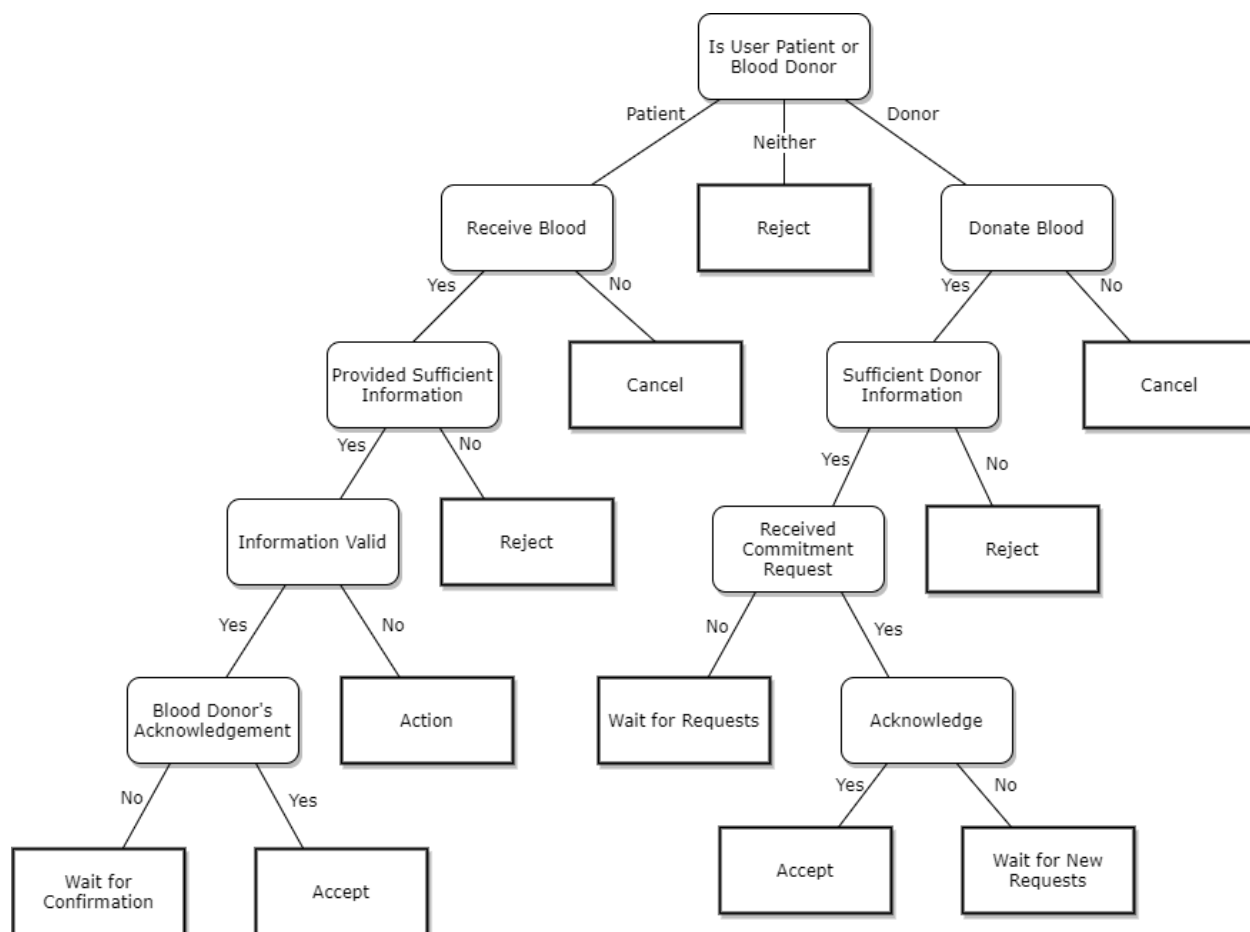


Figure-9: Decision Tree for "Blood Donation" feature

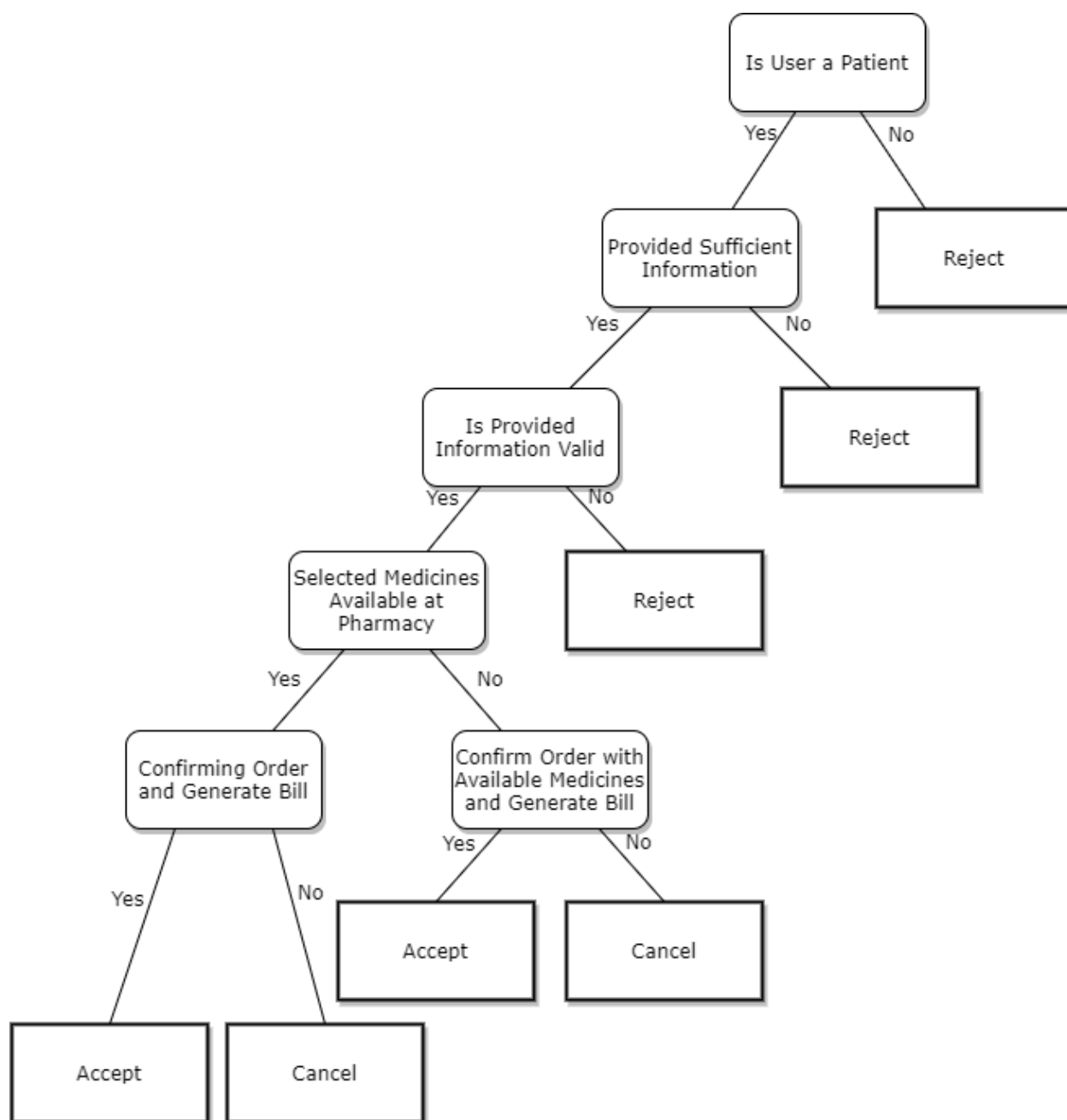


Figure-10: Decision Tree for "Home Delivery of Medicine" feature

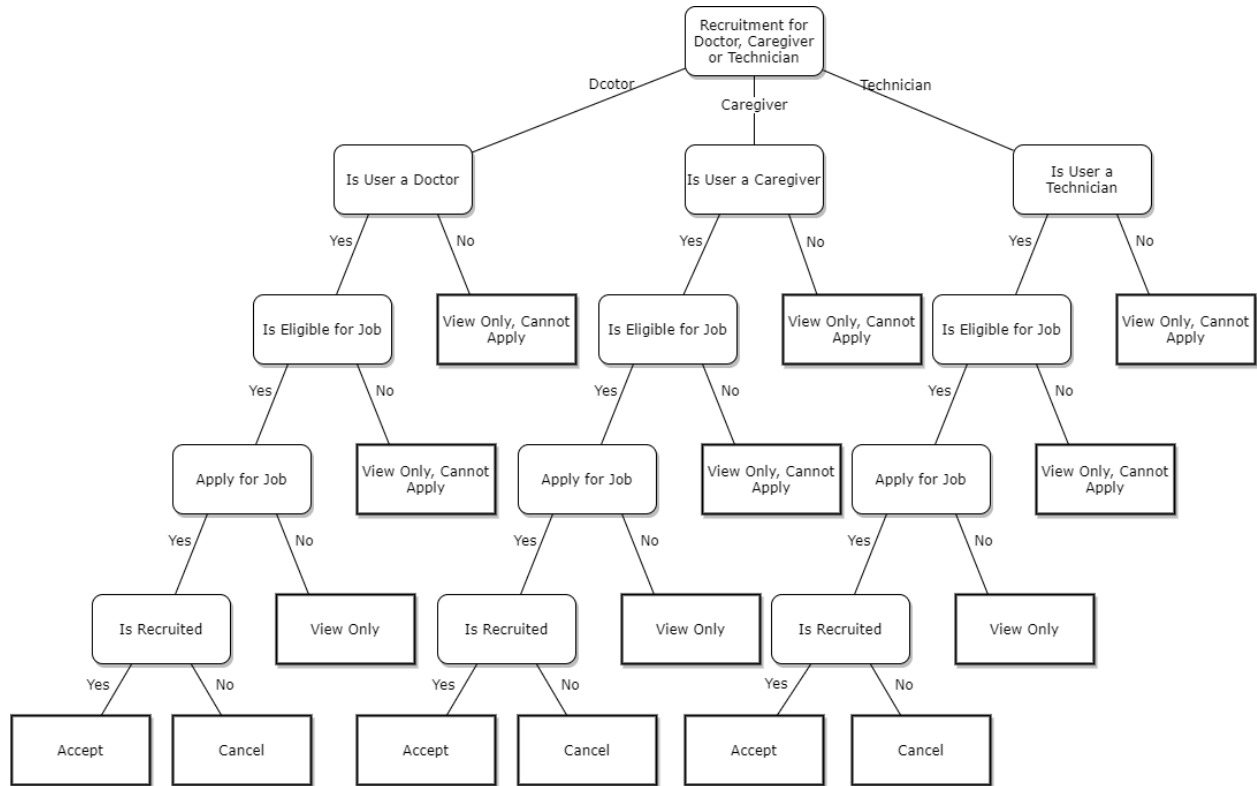


Figure-11: Decision Tree for “Recruitment” feature

Table-5: Create Prescription

Condition Stub	Action Stub			
Is User a doctor?	No	Yes	Yes	Yes
Provided valid information?		No	Yes	Yes
Valid Patient ID?			No	Yes
Reject	X	X	X	
Accept				X
Condition Entry	Action entry			

Table-6: Take Appointment

Condition Stub	Action Stub				
Is User a doctor?	No	Yes	Yes	Yes	Yes
Provided sufficient information?		No	Yes	Yes	Yes
Valid doctor information given?			No	Yes	Yes
Is the doctor available at the requested time?				No	Yes
Reject	X	X	X		
Cancel				X	
Accept					X

Table-8: Hire caregiver

Condition Stub	Action Stub					
Is the User a Patient?	No	Yes	Yes	Yes	Yes	Yes
Provided sufficient information?		No	Yes	Yes	Yes	Yes
Valid caregiver information given?			No	Yes	Yes	Yes
Is the caregiver occupied?				Yes	No	No
Caregiver's acknowledgement?					No	Yes
Reject	X	X	X	X		
Cancel					X	
Accept						X
Condition Entry	Action entry					

Cost and Benefit Analysis:

Cost and benefit analysis of existing system:

- **Cost:** The existing healthcare system has a cost approximately less. The main cost is for domain and hosting cost. There is no system which incorporates all our concepts about project medica. The domain cost is 3-10 USD and for hosting it needs 50-100 USD per year. For maintenance it costs 400-500 USD.
- **Benefit:** This kind of small system is easy to maintain and administration can handle the users database easily. Still it is suitable for the small system not for the universal healthcare system
- **Analysis:** The estimated cost benefit analysis shows that constructing and operating a system like this is less costly but has limitations like security, reliability and performance.

Cost and benefit analysis of candidate system:

- **Cost:** The domain cost is 3-10 USD. As our system will be hosted on the cloud it costs 80-160 USD. For maintenance it will cost 400-500 USD. Cloud database systems are more reliable than the normal servers.
- **Benefit:** As data will be on cloud there will be more increase in performance, system will be more reliable and data security will be ensured.
- **Analysis:** The estimated cost benefit analysis shows that constructing and operating a system like this is more costly but it will be suitable for the big scale system. We can make a scalable system with this and ensure more performance than existing systems.

5. Conclusions

The candidate system is to ease the workings of the health service providers as well as the service receivers. Doctors, care-givers and patients have to be benefited through the implementation of the system. Digitalization of health services has been a requirement for a long time and which can be fulfilled by the selected candidate system. Using this web-based application, the patients will easily access their prescriptions, reports, and be able to contact the doctors. Again the doctors and caregivers will be able to access their schedule and keep track of all the necessary dealings with the patients. Hospitals, too can maintain their formal regulations through this application. *Medica* provides an all-around service to people seeking and providing services in the field of healthcare.

6. Appendices

Appendix-A

A Universal Healthcare Service Management System

The information gathered from here will be used to develop a universal healthcare service management service for patients, doctors and everyone.

***While filling up the form, consider yourself as a hospital's administrative member.

It'll help us to collect data for our project. Thanks in advance.

Are you comfortable with paper-based info management of a hospital's healthcare? *

☐ Yes

☐ No

Would you prefer a digital health management system over a paper-based management approach? *

☐ Yes

☐ No

Do you have a digital health management system of your own?

- ☐ Of course
- ☐ No
- ☐ Will launch one soon
- ☐ Other...

How often do you sit for consultancy?

- ☐ 7 days a week
- ☐ 4-6 days a week
- ☐ 2-3 days a week
- ☐ Other...

How do you maintain and improve your digital system? *

- ☐ Checking whether data is properly handled
- ☐ Receiving user reviews and recommendations
- ☐ Development team comes to check once in a while
- ☐ We don't. It stays the way it is.
- ☐ Sorry. We don't have one

What steps have you taken to protect your system from malicious attacks? Please write your answer.

Short answer text

Does your system support creation of varieties of test reports? *

- ☐ Yes, for every categories of reports
- ☐ Not all. Some are created with a separate software.
- ☐ We rely on a separate software

How do you deliver a test report to a patient? *

- ☐ Patients come in-person
- ☐ Email
- ☐ Postal
- ☐ Other...

How do customers try to contact you for inquiries? *

- ☐ Email
- ☐ Phone
- ☐ ...

Do you ever disclose patient's personal data, including health records, to other hospitals or organizations for research purposes? *

- ☐ Yes
- ☐ No
- ☐ Maybe

Is there any means in your current existing system to preserve a patient's prescription, be it digital or a hard-copy? *

- ☐ Yes
- ☐ No

Do you think the doctors should check a patient's entire medical history instead of recent ones before prescribing? *

- ☐ Yes
- ☐ No
- ☐ No Need

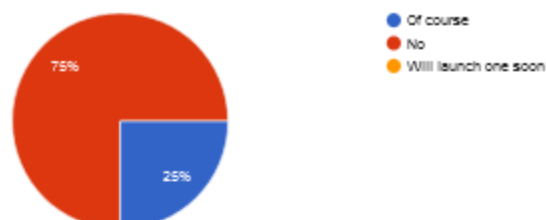
Do you use any application to regulate your everyday schedules? *

- ☐ Yes

Appendix-B

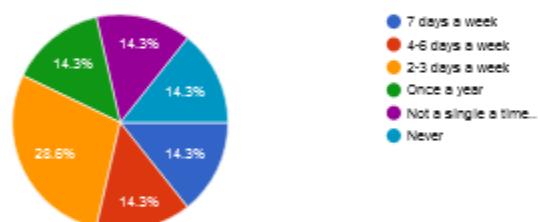
Do you have a digital health management system of your own?

8 responses



How often do you sit for consultancy?

7 responses



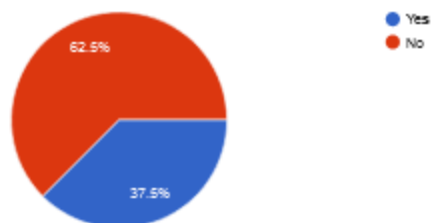
How do you maintain and improve your digital system?

8 responses



Are you comfortable with paper-based info management of a hospital's healthcare?

8 responses



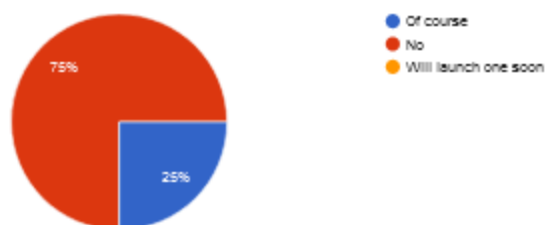
Would you prefer a digital health management system over a paper-based management approach?

8 responses



Do you have a digital health management system of your own?

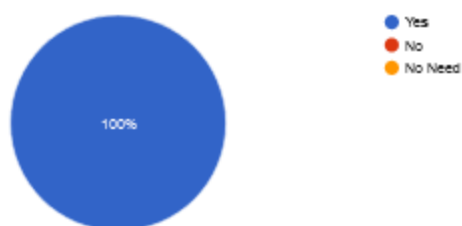
8 responses



Do you think the doctors should check a patient's entire medical history instead of recent ones before prescribing?



8 responses



Do you use any application to regulate your everyday schedules?

8 responses



Do you have any home delivery service for medicines?

8 responses

